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AI-Based Multimodal Framework for Shot-Level Analysis of Documentary-Style Micro-Lecture Videos

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***Abstract*—With the ongoing digital transformation of higher education, documentary-style micro-lecture videos have become an important medium for delivering content and shaping value orientation in ideological and political education. While prior studies have examined thematic content, narrative structures, or dissemination effects, micro-level structural modeling across visual, linguistic, and audio modalities remains limited. This study proposes an AI-based multimodal framework for shot-level analysis of such videos, integrating computer vision, natural language processing, and audio signal analysis within an AI-assisted multimodal framework validated through human cross-checking. The visual modality employs PySceneDetect for automatic shot boundary detection and OpenCV for representative frame extraction, followed by human cross-validated classification into five shot scales (C/MC/MS/LS/BG). The linguistic modality applies TF-IDF vectorization and K-Means clustering to the narration transcript, selecting K=5 based on Silhouette, Davies–Bouldin, and Calinski–Harabasz indices, and visualizing the results with t-SNE. The audio modality extracts per-shot mean pitch (F0) from voiced frames, providing a temporal profile of acoustic variation. The results from the visual, linguistic, and audio modalities are integrated into a multimodal rhythm diagram. This diagram presents shot-scale distribution, thematic clustering patterns, and pitch variation across the entire video. This visualization highlights the distinct structural features identified in each modality and provides a consolidated view of the documentary-style micro-lecture's composition.**

Keywords*—AI-Based video analysis; Multimodal analytics; Shot boundary detection; Shot scale classification; Documentary-style micro-lecture videos.

I. Introduction

The application of artificial intelligence (AI) in higher education is transforming approaches to analyzing the internal structure of instructional videos. Documentary-style micro-lecture videos—particularly in ideological and political education—combine pedagogical objectives with the production techniques of short-format documentaries and are characterized by deliberate narrative structuring, varied shot compositions, and integrated audio-visual elements. These videos exhibit multimodal organization: changes in visual framing, thematic progression in the narration, and fluctuations in acoustic features together form a temporal rhythm that shapes the presentation of information and values.

Multimodal methods have been applied in diverse contexts, for example, in TikTok-based tourism promotion, where verbal and visual resources construct sustainable imaginaries[1]. Some scholars have focused on VR-based learning, capturing temporal interactions between cognition, behavior, and emotion through physiological signals, motion, and verbalized monitoring[2]. Other research has examined participatory educational settings, where Digital Stories documented student experiences and social semiotic analysis highlighted varied forms of expression [3].

In recent years, multimodal and discourse-oriented approaches have been increasingly applied across a wide range of contexts, including documentaries, classroom interactions, social media platforms, tourism discourse, and cultural heritage experiences. These studies highlight how visual, verbal, and auditory resources intersect to construct narratives, identities, and affective engagements, thereby advancing our understanding of both educational practice and broader sociocultural communication. For instance, in documentary research, multimodal critical discourse analysis of the BBC *Seven Up* series has shown how visual, verbal, and audio modes jointly contribute to the diachronic construction of interviewee images, reflecting shifts in ideology and socio-political reality[4]. Multimodal analysis has also been applied in history learning with mobile technologies, where video data revealed how emotional engagement was fostered through multimodal layering of stimuli, digital artefacts, and changes in pace[5]. In simulation-based training, multimodal learning analytics has been integrated with qualitative frameworks such as DiCoT, where video, speech, and eye-tracking data were combined to extend traditional analyses and generate more automated interpretations of learner performance[6].

However, few studies have developed a reproducible, shot-level framework that systematically integrates visual, textual, and audio modalities in the analysis of educational videos. This paper introduces a multimodal signal analysis framework designed to address these gaps by modeling educational videos at the structural level. The proposed approach integrates three distinct modalities:

Visual–shot scale classification to capture the visual composition and framing dynamics.

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Textual–semantic clustering of transcripts to identify thematic progression.

Audio–pitch-based contour extraction to approximate emotional intensity variations.

By situating the work at the intersection of signal processing, image analysis, and educational media research, this study offers both a technical methodology and a conceptual perspective to understand and optimize educational video design. The findings had practical implications for automated video annotation, video editing support, and instructional media evaluation.

II. Review of Related Literature

Multimodal learning analytics (MMLA) and multimodal discourse analysis have become central to contemporary educational research. Prior studies can be clustered into three strands: (i) learning behavior and collaboration modeling, (ii) classroom interaction and feedback practices, and (iii) disciplinary knowledge and educational discourse.

A. Learning behavior and collaboration modeling

MMLA has been used to characterize students' behaviors and their relationships to outcomes. In animated programming classrooms, algorithmic analysis clustered learners into "stay active," "stay passive," and "shift-to-passive" profiles, which were associated with learning performance[7]. In authentic classrooms, the role of data quality has been examined, showing that collaboration-quality models are sensitive to attribute noise and that random forest and decision tree tend to be more robust than alternative classifiers under such noise[8]. Beyond single studies, a system architecture (SPARC) has been articulated to standardize, process, analyze, recognize, and conceptualize multimodal trace data for modeling real-time self-regulated learning, with an explicit bridge from theoretical SRL constructs to AI-assisted inference and intervention[9].

B. Classroom interaction and feedback practices

Multimodal perspectives have also informed teaching and feedback. Studies of screencast-based feedback have shown that the configuration of voice, text, and on-screen visual markers enriches the meaning potential of written comments, foregrounding the sociometrical design of feedback in initial teacher education[10]. Longitudinal, video-based analyses of touch-screen-supported collaboration have documented how researchers and teachers can co-inquire into micro-level interaction, highlighting embodied, material, and linguistic dimensions and reflecting on methodological challenges of such partnerships[11].

C. Disciplinary knowledge and educational discourse

At the level of curricular and public discourse, multimodal analyses have clarified how meanings are constructed across media. A large-scale textbook study (Grades 7–10; N=749 diagrams) shows that explanation diagrams characteristically employ narrative and temporal functions and annotated captions, complementing written language to support reasoning about dynamic processes and time-bounded sequences[12]. On social platforms, analyses of educational profiles and threads on Twitter detail the hypertextual complexity of discussions and the organizational role of multimodal resources in online educational debates[13].

Across these strands, prior research demonstrates the breadth of MMLA for profiling behavior, supporting feedback, and analyzing discourse. However, few studies provide a semi-automated, reproducible framework that structurally integrates visual, textual, and audio modalities at the shot level for educational video—particularly for Documentary-style micro-lectures. This study addresses the lack of structural multimodal analysis in existing research on university course videos, which often treat modalities in isolation or as simple overlays. Focusing on Documentary-Style Micro-Lecture Videos, it aims to (1) To characterize visual shot scales, textual themes, and audio rhythms separately and examine their potential associations in narrative construction without strict temporal alignment, and (2) To integrate these results into a unified rhythm diagram that supports the optimization of content organization and narrative presentation. By independently modeling each modality and then structurally integrating the outcomes, the study provides a new pathway for multimodal narrative structure modeling of educational videos.

III. Experimental Design and Overall Approach

This study analyzes a Documentary-Style Micro-Lecture Video, which differs from both classroom recordings and conventional documentaries through its refined structure, strong narrative drive, and visual diversity. Beyond conveying knowledge, such videos aim to construct a sense of political rituality and emotional resonance via multimodal coordination. To examine their structural rhythm and narrative mechanisms, we adopted a three-modality framework—visual, linguistic, and audio. Each modality is modeled independently: the visual modality captures shot-scale distribution and editing rhythm, the linguistic modality traces thematic units and semantic progression, and the audio modality reflects emotional intensity through acoustic variation. The results are then integrated within a unified narrative framework to reveal how the modalities complement one another in information delivery, emotional modulation, and ritual construction, and provides a systematic path for understanding the multimodal communication structure of Documentary-Style Micro-Lecture Videos.

A. Visual Modality Analysis: Shot-Scale Distribution and Narrative Rhythms

As the first component of the three-modality analysis, the visual modality directly shapes the spatial presentation and emotional tone of *Documentary-Style Micro-Lecture Videos*. Structurally, these videos rely on variations in shot scale to highlight focal points and establish rhythm, while background images and inserted static information reinforce paragraph boundaries and ritualistic effects. In this section, shot boundary detection combined with manually annotated shot-scale classification is used to construct a visual rhythm diagram of the entire video, through which the distributional characteristics and collaborative functions of different shot scales in narrative construction are examined.

- Shot Boundary Detection and Key Frame Extraction. The first step of visual analysis is shot boundary detection. Using Python and the open-source tool PySceneDetect, the video was segmented automatically based on inter-frame pixel variation, producing time points that define discrete visual units (shots). From

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each shot, a mid-frame was then extracted as a representative image via the OpenCV library, balancing duration, clarity, and representativeness. This process ensures consistency and reproducibility in subsequent shot-scale classification and provides a stable temporal basis for multimodal integration.

- Shot-Scale Labeling and Manual Identification. After shot segmentation and key-frame extraction, a five-category labeling system was established: **C (Close-up)** focusing on facial details, **MC (Medium Close-up)** showing the upper body, **MS (Medium Shot)** balancing half-body and context, **LS (Long Shot)** presenting full figures or wide views such as aerial or campus scenes, and **BG (Background)** frames without speakers, used for transitions with symbolic imagery. Representative frames were manually annotated and cross-validated by multiple coders to ensure consistency and accuracy.

Although shot-scale identification in this study involved manual cross-validation, the framework remains AI-assisted rather than fully manual. Automated modules, including PySceneDetect for boundary detection and OpenCV for frame extraction, constitute the primary computational pipeline, while human annotation functions as a reference benchmark to ensure classification reliability in the absence of a large-scale labeled dataset. This design balances automation and interpretability, enabling verifiable results under current data constraints.

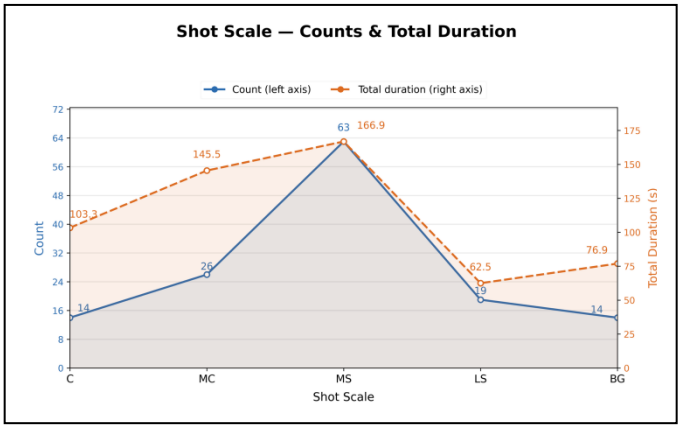


Fig. 1. Shot Scale — Counts & Total Duration

As shown in Figure 1, shot types varied significantly in both frequency and duration. Medium shots (MS) were most prevalent, with 63 shots totaling 167.4 seconds, providing visual stability and narrative support. Medium close-ups (MC, 26 shots, 146.2s) and close-ups (C, 14 shots, 103.5s) emphasized expressions, thematic focus, and emotional rendering. Long shots (LS, 19 shots, 63.4s) extended spatial scope, interweaving aerial and campus views to highlight institutional atmosphere and achievements, creating a "space-extension to narrative-convergence" effect. Background shots (BG, 14 shots, 76.8s) marked structural transitions, combining red motifs and titles to establish ritualized boundaries. Some BG shots also carried textual or static images introducing concepts, history, or symbols, enriching content delivery and providing narrative breathing space.

B. Linguistic Modality Analysis: Thematic Units and Narrative Progression

In Documentary-Style Micro-Lecture Videos, the linguistic modality serves as the primary vehicle for value transmission and narrative development. Using the complete transcript, semantic embedding and clustering methods were applied to identify thematic distributions and semantic relations, thereby revealing how the text organizes narrative rhythm. The analysis focuses not only on thematic segmentation but also on how these units advance logically and intensify emotionally to sustain narrative cohesion and impact.

- **Data Preparation and Preprocessing**. The dataset consists of a Chinese narration transcript of 2,054 characters, segmented into natural sentences as analysis units. Each unit was normalized (character formats, spacing, case), tokenized, and filtered using both a general stop-word list and a domain-specific high-frequency list, retaining primarily nouns and proper nouns to enhance thematic discrimination.

- **Text vectorization.** TF-IDF was applied to represent each unit as a sparse high-dimensional vector. To ensure reproducibility, all hyperparameters (e.g., minimum frequency threshold, maximum feature size) were fixed across the process.

- **Clustering and parameter selection.** K-Means was performed with $K \in \{3, 4, 5, 6\}$. For each K, multiple random initializations were run to mitigate sensitivity to starting points, and the best result was retained. Internal validity was then evaluated using three clustering metrics. Silhouette coefficient (the larger the better): measures intra-cluster cohesion and inter-cluster separation. Davies–Bouldin index (DB) (the smaller the better): measures inter-cluster overlap and intra-cluster scatter. Calinski–Harabasz index (CH) (the larger the better): measures the ratio of between-cluster variance to within-cluster variance. In the actual experiments, combining these three indices with interpretability assessment, **K=5** achieved the best balance of inter-cluster separation, intra-cluster consistency, and thematic clarity. In contrast, K=3/4 resulted in excessive merging of different themes; K=6 led to over-fragmentation of certain themes. Therefore, **K=5** was selected as the final number of clusters.

- **Visualization and Thematic Labeling.** t-SNE was used to embed high-dimensional vectors into a two-dimensional space for visualization. To ensure comparability across different K values, t-SNE hyperparameters and random seeds were fixed. Each cluster was marked with a distinct color, and its spatial distribution was outlined with a Gaussian-approximated ellipse. For each cluster, the top two keywords ranked by mean TF-IDF scores were selected as representative thematic labels, which were placed within the corresponding ellipse using an auto-avoidance strategy to prevent overlap with other labels. In the figure, nnn denotes the number of analysis units within a cluster, reflecting the natural imbalance in corpus distribution.

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Figure 2 visualizes clustering results for K = 3, 4, 5, and 6 in the same 2D space with internal validity indices. K = 5 shows clearer boundaries; K = 3/4 merges themes too broadly; K = 6 produces overlapping clusters. These results align with the metrics, supporting K = 5.

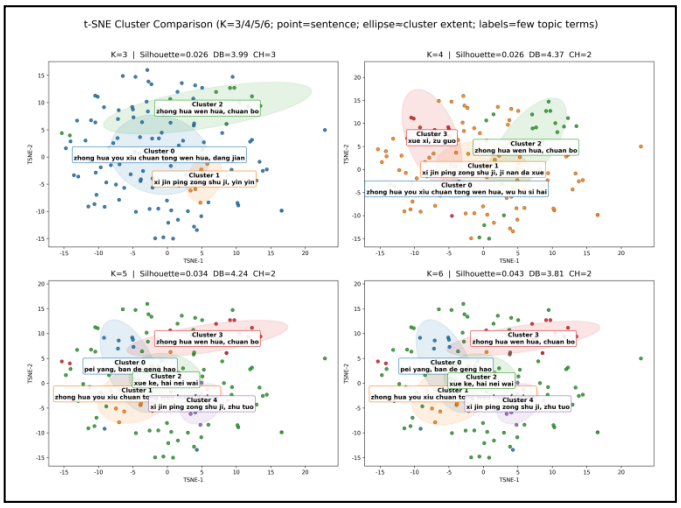


Fig. 2. t-SNE Cluster Comparison (K=3/4/5/6)

Figure 3 presents the final K = 5 solution, where five themes are distinctly distributed. Cluster sizes (n) indicate corpus imbalance: larger clusters cover broader themes, smaller ones represent more specific topics.

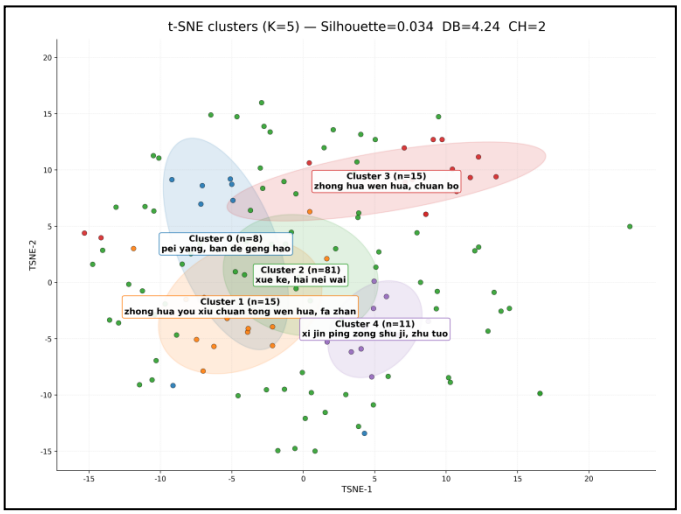


Fig. 3. Final t-SNE Clustering Result (K=5)

The K = 5 semantic clustering results indicated that the narration was organized into five distinct and logically progressive sections in constructing the red narrative: beginning with historical background and spiritual values, moving to a concise exposition of core ideas, then shifting to emblematic cases that embody emotion and conviction, extending into discussions that connect with contemporary realities, and finally concluding with a summary and call to action. This progression—from history, to principles, to cases, to reality, to exhortation—reflects a common value-transmission pathway in red narratives.

The thematic distribution and transitions shown in the clustering visualization further suggest that the text achieves the characteristic rhythm of red storytelling, combining emotional buildup with ideological elevation. Each section maintains semantic distinction while also resonating with others, reinforcing both emotional progression and logical coherence. This structure enables the narrative to move audiences emotionally while persuading them intellectually, demonstrating a high level of narrative maturity and communicative effectiveness.

C. Audio Modality Analysis: Acoustic Variation and Emotional Intensity

The audio modality captured the speaker's strategies of emotional regulation and auditory guidance through pitch fluctuations and rhythmic variation. In this section, the mean pitch of each analysis unit is extracted to trace acoustic trajectories and examine their role in emphasizing key information, creating emotional peaks, and providing transitional buffers. Combined with visual and linguistic features, the audio analysis offers auditory evidence for understanding the multimodal narrative coordination of Documentary-Style Micro-Lecture Videos.

- Audio Data Extraction and Preprocessing. After separating the course video, the audio stream was extracted as mono-channel WAV (sampling rate 22,050 Hz). To preserve vocal details and background characteristics, only amplitude normalization was applied, without denoising or compression.
- Shot-Level Pitch Extraction. Based on identified shot boundaries and IDs, corresponding audio segments were clipped. Frame-level pitch values (Hz) were extracted for each segment, and the mean of all valid pitches was calculated as the representative pitch indicator. Pitch extraction was implemented using the librosa.pyin algorithm in Python, which applies a probabilistic variant of the YIN fundamental-frequency estimation method. This approach provides reliable monophonic pitch tracking across voiced frames and ensures methodological reproducibility through fixed frame length (2048 samples) and hop length (512 samples) parameters. Compared with full-track analysis, this finer granularity captures subtle prosodic variations across shots, characterizing shifts in vocal intensity and emotional transitions.

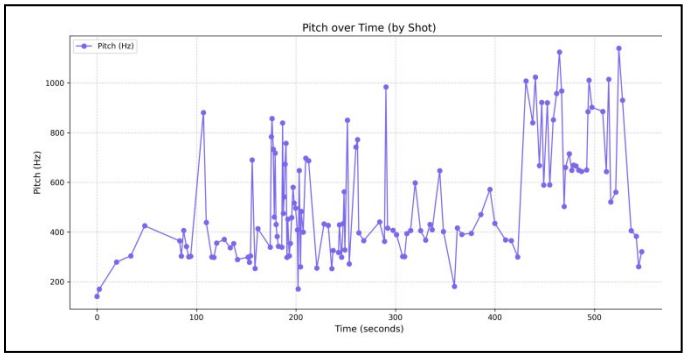


Fig. 4. Pitch over Time (by shot)

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- Rhythmic Features and Visual Correlation. Figure 4 illustrates the temporal trajectory of mean pitch values across shots. Peaks coincide with heightened emotional delivery, while explanatory or background segments display flatter contours. Comparison with shot-switching frequency revealed that certain pitch peaks aligned with high-frequency visual transitions, suggesting potential cross-modal coordination at key narrative junctures.
- Methodological Robustness and Limitations. To ensure robustness, fixed frame length and overlap parameters were applied during pitch extraction, and outliers (e.g., extreme frequencies outside the normal human vocal range) were removed. However, as segmentation was based solely on shot-level timelines without direct alignment to the textual modality, the results primarily capture audio–visual rhythmic relations rather than cross-modal semantic synchrony. Future work could extend this approach to joint three-modality rhythm analysis when precise textual timestamps are available.

To establish cross-modal comparability, sentence-level linguistic units were temporally aligned with shot-level segments through approximate timestamp mapping derived from narration continuity. When a sentence spanned multiple shots, it was linked to the shot covering the largest temporal proportion of its duration. This soft alignment strategy preserves the temporal rhythmic correspondence between modalities without enforcing frame-level synchronization. The resulting integration thus reflects structural rhythm alignment rather than extremely micro-level semantic co-occurrence, consistent with the analytical scope of this study.

D. Multimodal Integration Analysis

Although the three modalities in this study were analyzed independently at the methodological level, they demonstrated strong complementarity at the narrative level. As shown in Figure 5, a "visual–linguistic–audio" multimodal narrative analysis framework was constructed, linking the data layer, tool layer, narrative analysis layer, meaning layer, and integration layer in sequence to illustrate the complementary mechanisms and integration pathways of multimodal information in narrative progression.

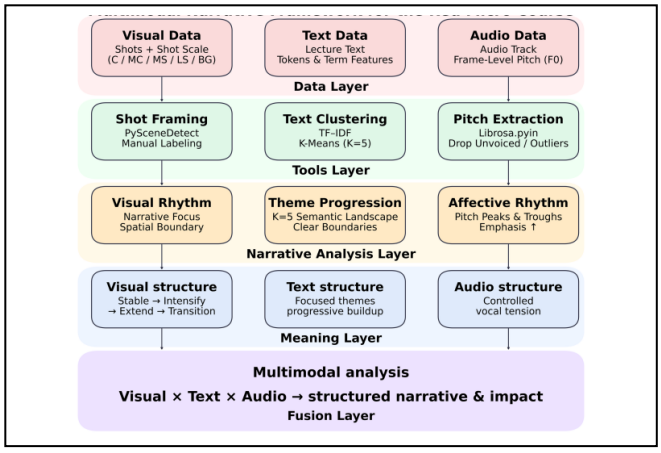


Fig. 5. Multimodal Narrative Framework of Documentary-Style Micro-Lecture Videos

The visual modality (shot-scale distribution) revealed rhythmic features of spatial presentation and narrative stability. Frequent use of medium shots (MS) establishes a stable visual baseline, while medium close-ups (MC) and close-ups (C) intensify expressions and narrative focus. Long shots (LS) and background shots (BG) support transitions and extend spatial scope, providing continuity and framing.

The linguistic modality (semantic clustering) demonstrated the thematic trajectory of content progression. The clustering distribution reflects a layered structure: from macro-level policy and ideological guidance to disciplinary development and cultural transmission, and finally to exemplary figures and value demonstration. These thematic progressions correspond with shifts in visual modality: for example, when the narration turns to themes of cultural communication and ideological guidance, they are often paired with stable medium shots and expressive close-ups, producing narrative peaks.

The audio modality (pitch contour) traced emotional fluctuation and prosodic rhythm throughout the narration. High-frequency segments aligned closely with key thematic clusters in the linguistic modality and often coincided with tightened shot scales in the visual modality, thereby creating dual reinforcement of emotion across auditory and visual channels.

This multimodal complementary mechanism enabled Documentary-Style Micro-Lecture Videos not only to present a coherent logical chain in information delivery, but also to surpass traditional courses or conventional documentaries in emotional appeal and narrative immersion, embodying the organic integration of political communication and audiovisual language.

IV. Conclusion

This study took Documentary-Style Micro-Lecture Videos as its research object. Building on a priori analysis of their narrative features and communicative logic, it proposed and applied a multimodal collaborative analysis framework tailored to this type of narrative content. By examining visual, linguistic, and audio modalities through structured rhythm analysis and cross-modal comparison, the study revealed underlying patterns in content organization, emotional transmission, and audience perception. The contributions of this work lie in its focus on the thematic attributes and political-communication context of Documentary-Style Micro-Lecture Videos while embedding multimodal methods into narrative inquiry, thereby aligning methodological construction with content characteristics; in demonstrating that the integration of shot-scale distribution, semantic clustering of transcripts, and pitch contour analysis can uncover rhythm coupling and emotional resonance across modalities, revealing the multimodal mechanisms that shape narrative climax and intensification; and in introducing the multimodal rhythm diagram as an integrative tool to map heterogeneous features onto a unified structural hierarchy, thereby enhancing visualization and interpretive capacity. Overall, this research not only provided a practical pathway for content analysis and narrative optimization of Documentary-Style Micro-Lecture Videos but also offered a transferable framework for multimodal studies of educational and political communication videos. Future work will explore the mapping between narrative structure and emotional response, the

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mechanisms of inter-modal interaction, and cross-context transferability, with the aim of producing research outcomes that offer stronger predictive and explanatory power in cognitive communication and narrative design.

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